



The University of Georgia

Center for Applied Isotope Studies

RADIOCARBON ANALYSIS REPORT

January 9, 2025

W. Christopher Carleton/Ilgner
Max Planck Institute of Geoanthropology
Kahlaische Strasse 10
Jena, 07745
Germany

Dear Dr. Carleton,

Enclosed please find the results of ^{14}C Radiocarbon analyses and Stable Isotope Ratio $\delta^{13}\text{C}$ analyses for the samples received by our laboratory on November 21, 2024.

The bulk sediment samples after root separation by sieving through 125 μm nylon screen and following acid treatment with 1N hydrochloric acid at 80°C for 1 hour were washed with deionized water using centrifuge to pH=4-5, then they were dried at 105°C. The dry samples was combusted at 900°C in the evacuated sealed quartz ampoule in the presents of CuO.

The charcoal sample was treated following the acid/alkali/acid (AAA) protocol involving three steps: (1) an acid treatment (1N HCl at 80°C for 1 hour) to remove secondary carbonates and acid-soluble compounds; (2) an alkali (NaOH) treatment; and (3) a second acid treatment (HCl) to remove atmospheric CO_2 . Sample was thoroughly rinsed with deionized water between each step, and the pretreated sample was dried at 105°C. For accelerator mass spectrometry analysis the cleaned samples were combusted at 900°C in evacuated / sealed ampoules in the presence of CuO

The resulting carbon dioxide was cryogenically purified from the other reaction products and catalytically converted to graphite using the method of Vogel et al. (1984) Nuclear Instruments and Methods in Physics Research B5, 289-293.

Graphite $^{14}\text{C}/^{13}\text{C}$ ratios were measured using the CAIS 0.5 MeV accelerator mass spectrometer. The sample ratios were compared to the ratio measured from the Oxalic Acid I (NBS SRM 4990). The sample $^{13}\text{C}/^{12}\text{C}$ ratios were measured separately using a stable isotope ratio mass spectrometer and expressed as $\delta^{13}\text{C}$ with respect to PDB, with an error of less than 0.1‰. The quoted uncalibrated dates have been given in radiocarbon years before 1950 (years BP), using the ^{14}C half-life of 5568 years. The error is quoted as one standard deviation and reflects both statistical and experimental errors. The date has been corrected for isotope fractionation

Sincerely,

Alexander Cherkinsky, Ph.D.
Senior Research Scientist

UGAMS #	Sample ID	Material	$\delta^{13}\text{C}, \text{‰}$	^{14}C age years, BP	$\pm$	pMC	$\pm$
73631	KKPP24_1_10	sediment	-24.83	1120	20	86.95	0.21
73632	KKPP24_1_50	sediment	-26.35	4340	25	58.27	0.17
73633	KKPP24_1_100	sediment	-26.63	3090	20	68.10	0.18
73634	KKPP24_2_10	sediment	-23.75	1530	20	82.67	0.21
73635	KKPP24_2_40	sediment	-27.13	4130	20	59.78	0.17
73636	KKPP24_2_80	sediment	-28.43	7880	30	37.50	0.13
73637	KKPP24_3_10	sediment	-19.71	950	20	88.87	0.21
73638	KKPP24_3_50	sediment	-23.85	2000	20	77.92	0.19
73639	KKPP24_3_90	sediment	-25.43	3850	20	61.90	0.17
73640	KKPP24_4_10	sediment	-22.02	490	20	94.13	0.23
73641	KKPP24_4_50	sediment	-25.06	2820	30	70.39	0.28
73642	KKPP24_4_90	sediment	-27.94	8290	20	35.64	0.12
73643	KKPP24_5_10	sediment	-23.08	modern	20	120.11	0.33
73644	KKPP24_5_40	sediment	-24.72	3050	20	68.43	0.18
73645	KKPP24_5_60	sediment	-25.05	4110	25	59.93	0.18
73646	KKPP24_6_10	sediment	-22.25	950	20	88.83	0.23
73647	KKPP24_6_60	sediment	-25.85	2370	20	74.41	0.19
73648	KKPP24_6_120	sediment	-25.07	6510	35	44.46	0.19
73649	KKPP24_7_10	sediment	-19.97	550	20	93.42	0.23
73650	KKPP24_7_50	sediment	-24.00	3250	20	66.75	0.19
73651	KKPP24_7_80	sediment	-24.11	3420	20	65.34	0.18
73652	KKPP24_8_10	sediment	-20.53	960	20	88.74	0.23
73653	KKPP24_8_40	sediment	-22.83	2020	20	77.76	0.19
73654	KKPP24_8_70	sediment	-27.04	8600	20	34.28	0.14
73655	KKPP24_4_70	charcoal	-26.84	1180	20	86.32	0.21
73656	KKPP24_7_70	charcoal	-26.93	1140	20	86.76	0.21
73657	KKPP24_8_30	charcoal	-28.55	270	20	96.66	0.25
73658	KKPP24_7_60	charcoal	-25.63	1230	20	85.78	0.20
73659	KKPP24_7_40	charcoal	-28.72	1210	20	86.04	0.22
73660	KKPP24_6_80	charcoal	-23.13	1270	25	85.37	0.26
73661	KKPP24_7_70a	charcoal	-27.29	1210	20	85.99	0.21
73662	KKPP24_3_10	charcoal	-28.20	860	20	89.88	0.22
73663	KKPP24_5_50	charcoal	-26.62	1240	20	85.65	0.24
73664	KKPP24_3_60	charcoal	-26.16	870	20	89.69	0.21
73665	KKPP24_3_68	charcoal	-26.70	940	20	88.94	0.23
73666	KKPP24_2_10	charcoal	-29.62	modern		119.43	0.27
73667	KKPP24_6_10	charcoal	-25.80	100	20	98.76	0.24
73668	KKPP24_5_30	charcoal	-25.49	1130	20	86.85	0.21